Quantum Wave Packet Scattering off a Schwarzschild Black Hole

A 3D Split-Operator Fourier Simulation Study

This report documents a fully three-dimensional numerical simulation of a non-relativistic quantum wave packet propagating in the presence of a Schwarzschild-inspired central potential. The wavefunction is evolved using a second-order split-operator Fourier method on a Cartesian grid, with complex absorbing boundary layers applied at every face of the simulation domain. Capture, reflection, and transmission probabilities are computed automatically by partitioning the computational domain relative to the event horizon and the initial direction of motion.

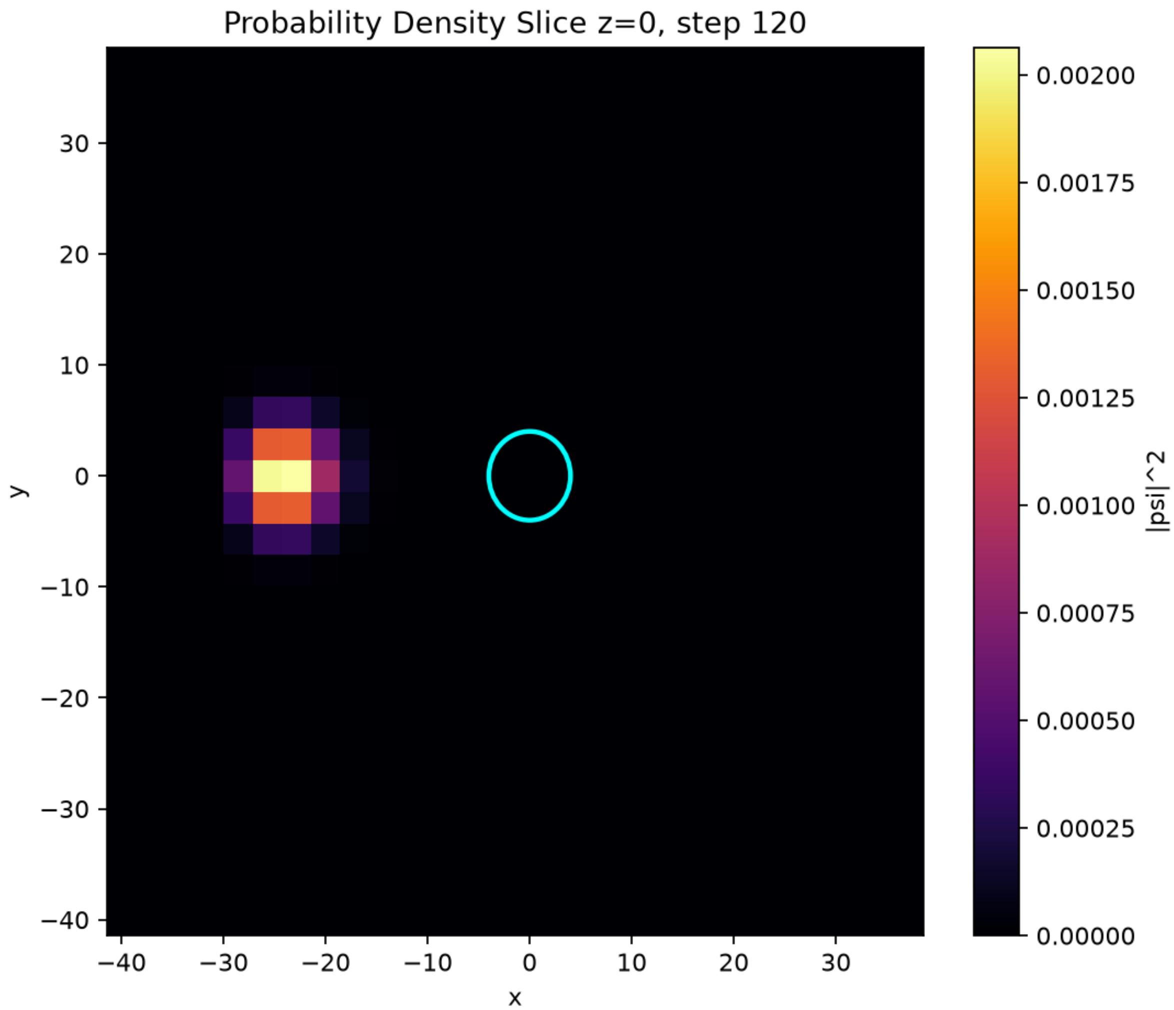
1. Simulation Configuration

Parameter	Value
Grid resolution	28 x 28 x 28
Domain size	80.0 x 80.0 x 80.0
Schwarzschild radius	4.0
Potential strength	1.0
Absorber width / strength	12.0 / 40.0
Initial packet center	(-25.0, 0.0, 0.0)
Initial momentum	(3.0, 0.0, 0.0)
Packet width (sigma)	3.0
Time step	0.01
Total steps	120

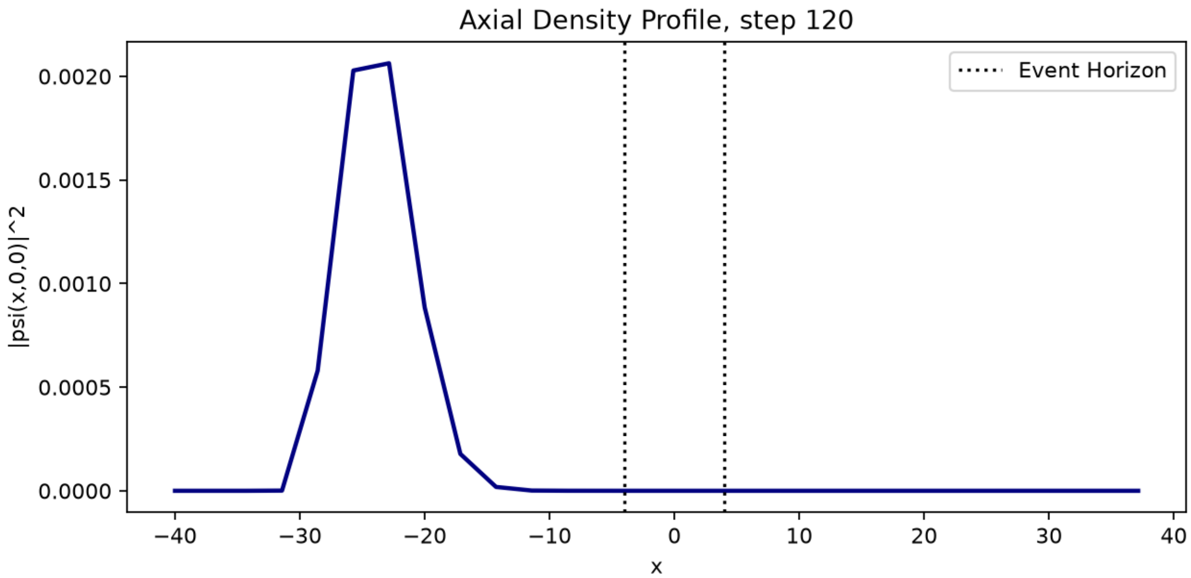
2. Baseline Run Results

Quantity	Value
time	1.200000
norm	0.933729
mean_x	-22.215275
mean_y	-0.000000
mean_z	-0.000000
mean_r	22.571226
kinetic_energy	0.333208
potential_energy	-0.047299
total_energy	0.285909
captured_probability	0.000014
reflected_probability	0.423302
transmitted_probability	0.510413

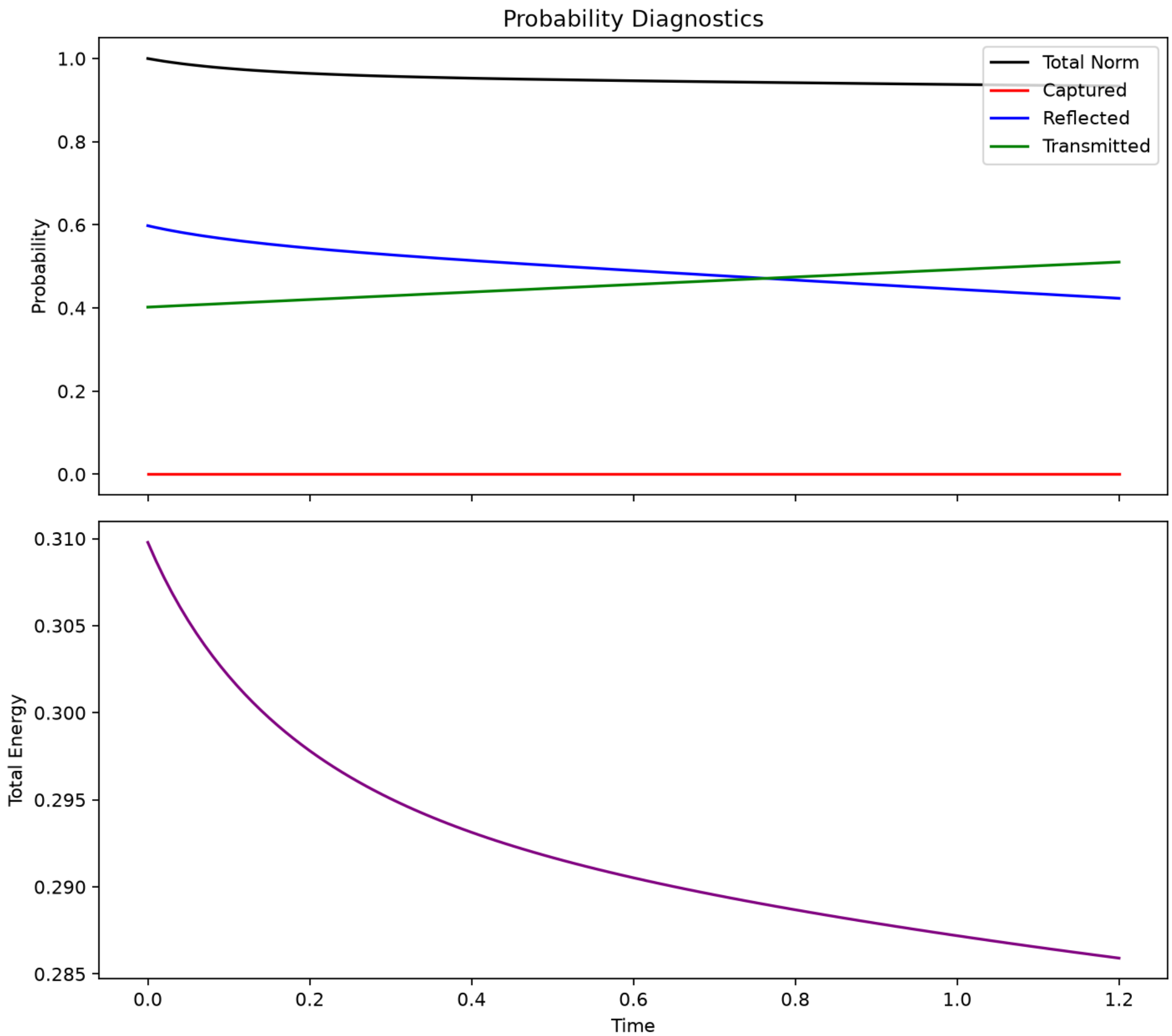
Probability Density Slice (z = 0)



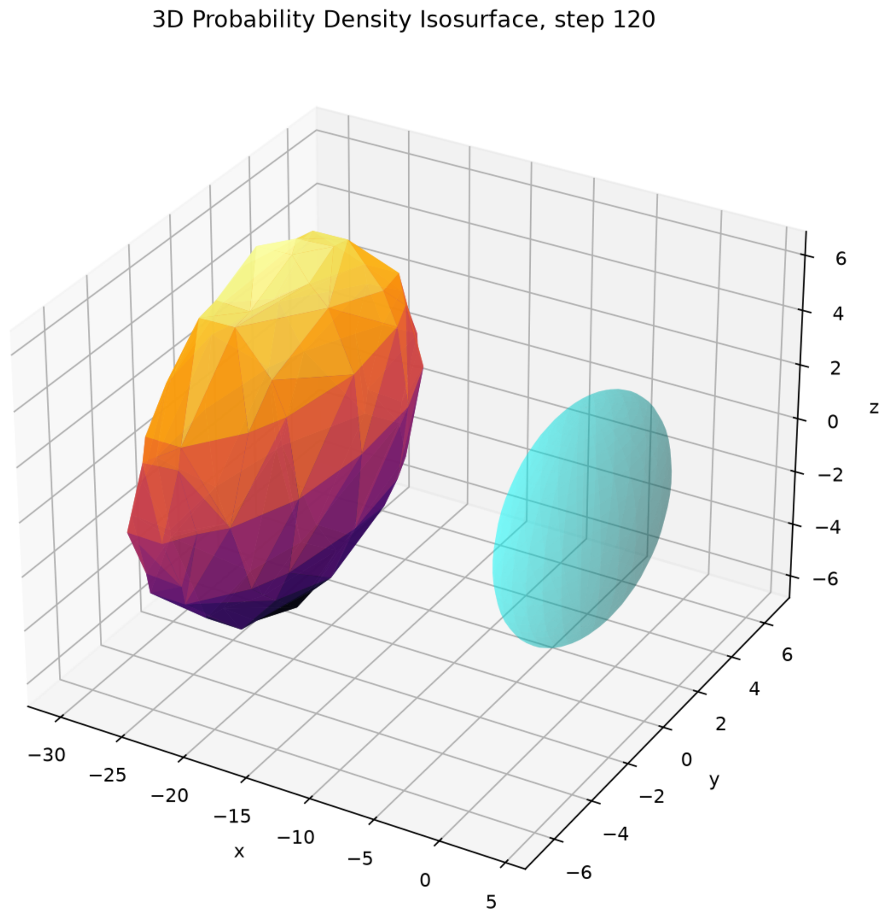
Axial Density Profile



Probability and Energy Diagnostics



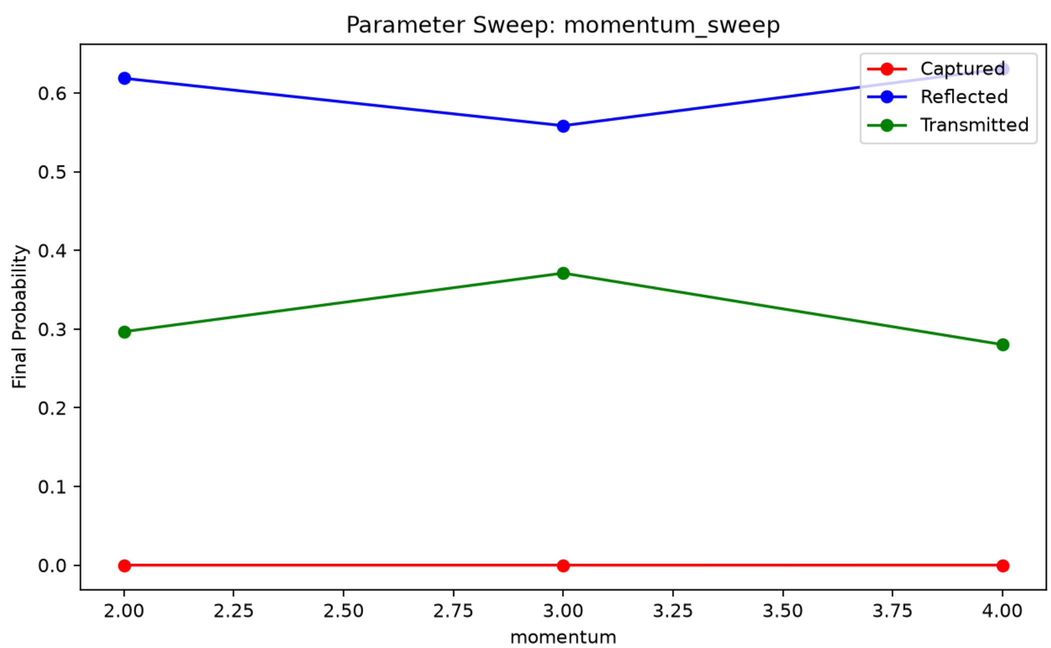
3D Probability Density Isosurface



3. Parameter Sweeps

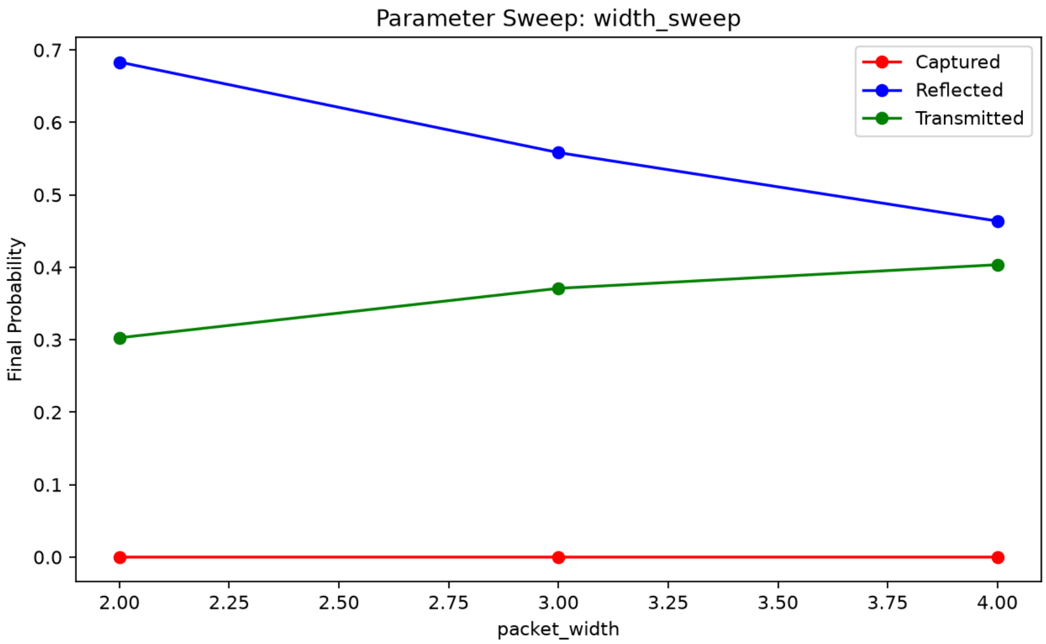
Momentum Sweep

Momentum	Norm	Captured	Reflected	Transmitted	Total Energy
2.0000	0.9153	0.0000	0.6186	0.2966	0.1202
3.0000	0.9294	0.0000	0.5584	0.3710	0.1100
4.0000	0.9110	0.0000	0.6308	0.2802	0.4506



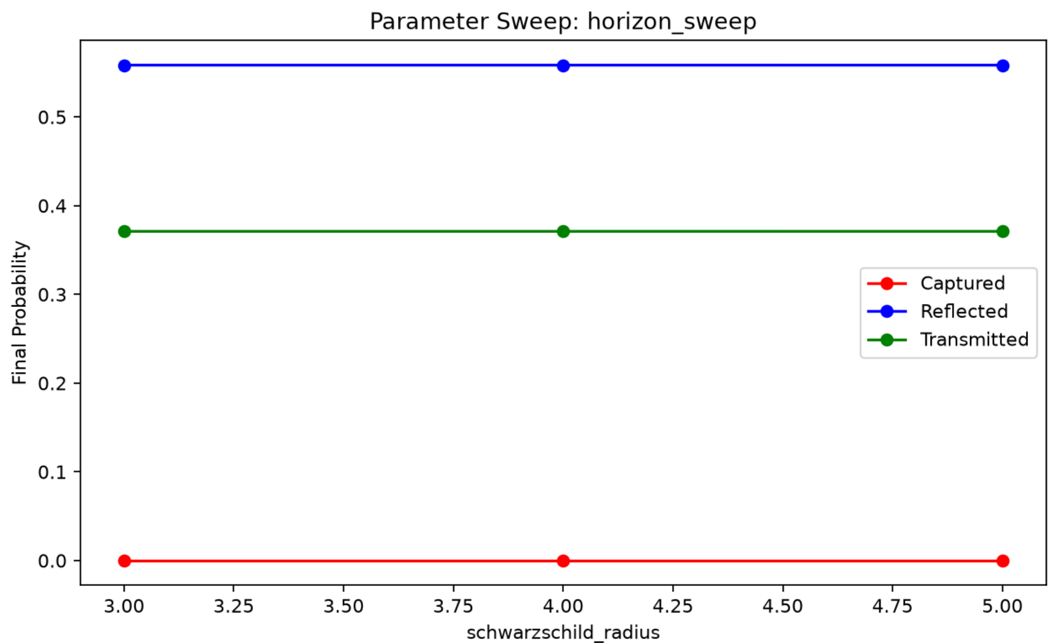
Width Sweep

Width	Norm	Captured	Reflected	Transmitted	Total Energy
2.0000	0.9861	0.0000	0.6833	0.3028	0.1662
3.0000	0.9294	0.0000	0.5584	0.3710	0.1100
4.0000	0.8675	0.0000	0.4639	0.4036	0.0865



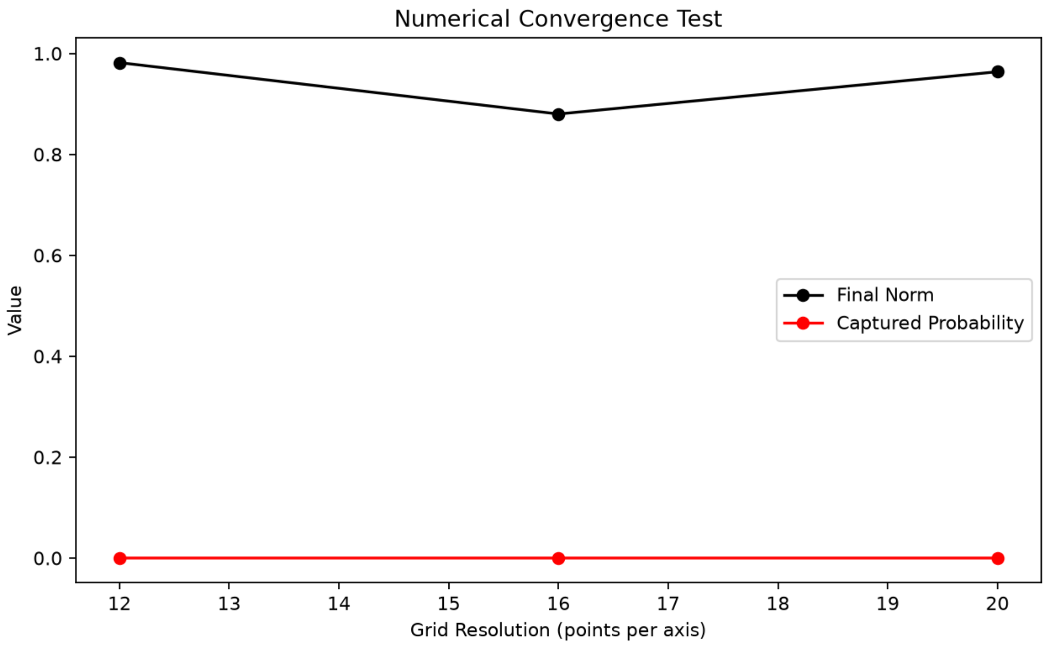
Horizon Sweep

Horizon Rs	Norm	Captured	Reflected	Transmitted	Total Energy
3.0000	0.9294	0.0000	0.5584	0.3710	0.1121
4.0000	0.9294	0.0000	0.5584	0.3710	0.1100
5.0000	0.9294	0.0000	0.5584	0.3710	0.1077



4. Numerical Convergence Test

Resolution	Norm	Captured	Reflected	Transmitted	Total Energy
12	0.9818	0.0000	0.7500	0.2318	0.0123
16	0.8800	0.0000	0.6990	0.1810	0.0725
20	0.9637	0.0000	0.3344	0.6292	0.0096



5. Summary

Under the baseline configuration, the simulation evolved a Gaussian wave packet with initial momentum (3.0, 0.0, 0.0) toward a Schwarzschild-inspired potential of horizon radius 4.0. The final captured probability was 0.000014, the reflected probability was 0.423302, and the transmitted probability was 0.510413. The parameter sweeps and resolution convergence test above characterize the sensitivity of these outcomes to the physical and numerical parameters of the model.